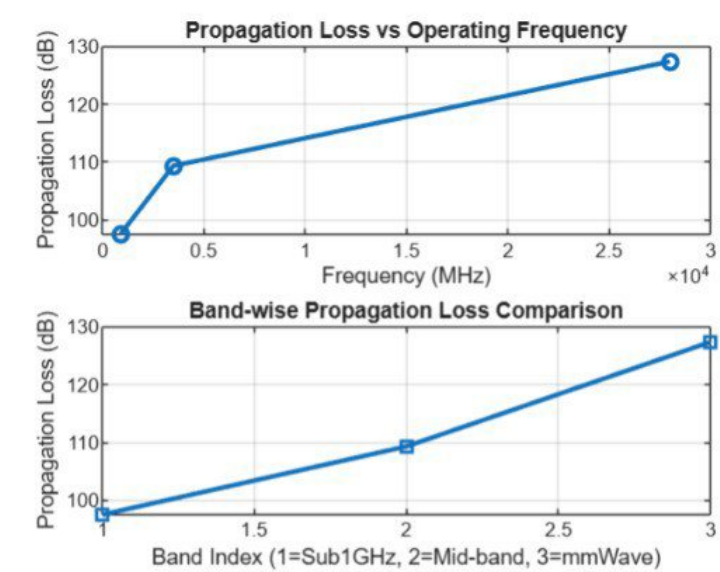
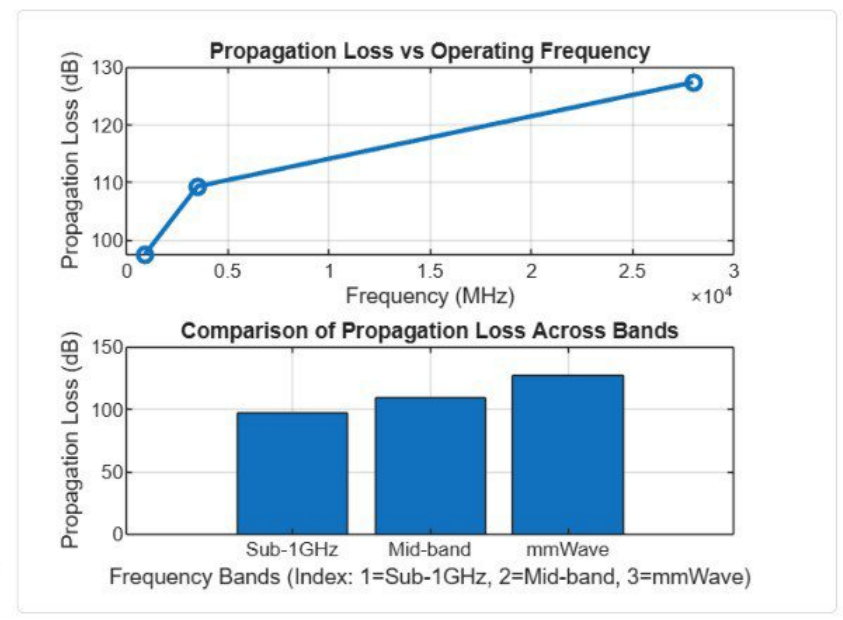


```
1 clc; clear; close all;
2 % Frequency bands (MHz)
3 f = [900 3500 28000];
4 % Distance (km)
5 d = 2;
6 % Propagation Loss (FSPL)
7 PL = 32.44 + 20*log10(f) + 20*log10(d);
8 figure
9 % ----- Graph 1: Simple Line Plot (SAFE) -----
10 subplot(2,1,1)
11 plot(f, PL, '-o', 'LineWidth', 2)
12 title('Propagation Loss vs Operating Frequency')
13 xlabel('Frequency (MHz)')
14 ylabel('Propagation Loss (dB)')
15 grid on
16 % ----- Graph 2: Safe Comparison Plot -----
17 subplot(2,1,2)
18 plot(1:3, PL, '-s', 'LineWidth', 2)
19 title('Band-wise Propagation Loss Comparison')
```



```

1 clc; clear; close all;
2 d = 2; % fixed distance (km)
3
4 f = [900 3500 28000]; % MHz
5 PL = 32.44 + 20*log10(f) + 20*log10(d);
6 figure
7 % ----- Graph 1: Frequency vs Loss -----
8 subplot(2,1,1)
9 plot(f,PL,'-o','LineWidth',2)
10 title('Propagation Loss vs Operating Frequency')
11 xlabel('Frequency (MHz)')
12 ylabel('Propagation Loss (dB)')
13 grid on
14 % ----- Graph 2: Band Comparison -----
15 subplot(2,1,2)
16 bar(PL)
17 title('Comparison of Propagation Loss Across Bands')
18 xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave)')
19 ylabel('Propagation Loss (dB)')
20 xticklabels({'Sub-1GHz','Mid-band','mmWave'})
21 grid on
    
```




```

1 % Frequency bands (MHz)
2 f = [900 3500 28000];
3 % Distance (km)
4 d = 2;
5 % Propagation Loss (FSPL)
6 PL = 32.44 + 20*log10(f) + 20*log10(d);
7 figure
8 % ----- Graph 1: Simple Line (d = 2 AFE) -----
9 subplot(2,1,1)
10 plot(f, PL, '-o', 'LineWidth', 2)
11 title('Propagation Loss vs Operating Frequency')
12 xlabel('Frequency (MHz)')
13 ylabel('Propagation Loss (dB)')
14 grid on
15 % ----- Graph 2: Safe Comparison Plot -----
16 subplot(2,1,2)
17 plot(1:3, PL, '-s', 'LineWidth', 2)
18 title('Band-wise Propagation Loss Comparison')
19 xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')
20 ylabel('Propagation Loss (dB)')
21 grid on
22

```

